

KijaniKiosk Capstone — Project Scope

Track A — Infrastructure-First | Controlled staging-to-production delivery | Rerimoi Tito | 2026

Problem Statement

KijaniKiosk needs a repeatable delivery path that separates staging from production and prevents an unvalidated change from being promoted blindly. This capstone addresses the gap by provisioning an isolated staging environment, externalizing configuration, validating application health automatically, testing deliberate failure and recovery, and placing explicit human approval before production promotion.

In Scope

- Terraform provisioning of the isolated **kijani-staging** namespace.
- Ansible management of staging application configuration through a Kubernetes ConfigMap.
- Kubernetes deployment and Service for **kk-payments** with three staging replicas.
- Jenkins delivery workflow covering staging deployment, rollout validation, smoke testing and manual production approval.
- Health monitoring plus deliberate failure/recovery validation.
- Operational documentation, test evidence, AI governance and peer-feedback records.

Out of Scope

- Application feature development or payment business-logic changes.
- Real customer traffic and production-scale performance validation.
- Multi-region or multi-cluster deployment.
- Enterprise-scale observability, managed database implementation and full cloud secret-management integration.
- Unattended automatic production promotion; production remains behind explicit human approval.

Success Criteria

Area	Success condition
Infrastructure	Staging namespace is reproducibly provisioned with Terraform and expected environment labels.
Configuration	Staging configuration is externalized and applied through Ansible.
Runtime	kk-payments reaches 3/3 ready replicas and /health returns {"status":"UP"}.
Failure handling	A deliberately broken image is detected as a failed staging deployment; restoring the known-good image returns service to healthy state.
Promotion	Production promotion is blocked until staging validation succeeds and explicit approval is given.

Architecture

Developer changes flow through GitHub and Jenkins. Terraform establishes the staging boundary; Ansible supplies environment configuration; Kubernetes runs kk-payments and its Service; rollout, readiness and smoke tests provide release evidence. A successful staging result leads to an explicit approval gate before production promotion.

Evidence: GitHub repository, staging smoke-test output, deliberate ImagePullBackOff failure/recovery, Jenkins approval gate, runbook, test plan and AI governance log.